

Atividade sobre análise de mutantes

Executei o teste de mutação com o Pitest na camada de serviços do projeto para avaliar se a nossa suíte de testes realmente consegue pegar erros caso o código mude no futuro. Tivemos um problema inicial de compatibilidade entre a versão do JUnit 5 do Spring Boot e o launcher do Pitest, mas resolvi alinhando as dependências no pom.xml. Com isso, o ambiente rodou perfeitamente.

```
=====
- Timings
=====
> pre-scan for mutations : < 1 second
> scan classpath : < 1 second
> coverage and dependency analysis : 4 seconds
> build mutation tests : < 1 second
> run mutation analysis : 4 seconds
-----
> Total   : 9 seconds
-----

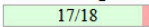
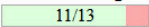
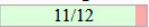
- Statistics
=====
>> Line Coverage (for mutated classes only): 17/18 (94%)
>> Generated 13 mutations Killed 10 (77%)
>> Mutations with no coverage 1. Test strength 83%
>> Ran 20 tests (1.54 tests per mutation)
Enhanced functionality available at https://www.arcmutate.com/

!! The following issues were detected during the run !!
Project uses Spring, but the Arcmutate Spring plugin is not present (https://docs.arcmutate.com/docs/sp)
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 12.314 s
[INFO] Finished at: 2026-06-10T19:07:05-03:00
[INFO] -----
19:07:05 PIT >> INFO : Completed in 9 seconds
```

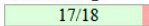
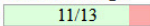
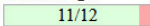
Resultados obtidos:

Pit Test Coverage Report

Project Summary

Number of Classes	Line Coverage	Mutation Coverage	Test Strength
1	94%  17/18	85%  11/13	92%  11/12

Breakdown by Package

Name	Number of Classes	Line Coverage	Mutation Coverage	Test Strength
br.org.edu.ifrn .LojaCarro.services	1	94%  17/18	85%  11/13	92%  11/12

Report generated by [PIT](#) 1.15.3

Enhanced functionality available at arcmutate.com

O resultado de 92% de força mostra que a nossa base de testes é muito confiável e protege bem a regra de negócio. Para chegar a 100%, o próximo passo é apenas criar testes para cobrir o valor de borda e o método que ficou sem cobertura.